

Literature Review

1) (Tyler et al., 2024)

(Tyler et al., 2024) conducted a scoping review of 29 studies published between 2018 and 2023 examining the application of artificial intelligence (AI) and machine learning in emergency department triage. The review synthesised studies that applied supervised and ensemble machine learning models to predict clinical outcomes, including hospital admission, mortality risk, disease severity, and resource utilisation. Across the included studies, ensemble methods such as XGBoost were frequently reported to outperform traditional triage systems such as the Emergency Severity Index (ESI) in predictive accuracy. Despite these findings, most studies were retrospective and single-centre in design, with limited external and prospective validation, reducing confidence in real-world clinical translation.

2) (Da'Costa et al., 2025)

(Da'Costa et al., 2025) conducted a literature review of AI-driven triage systems designed to address emergency department overcrowding, delayed patient assessment, and inconsistency in triage decision-making. The review examined applications of machine learning and natural language processing models aimed at improving patient prioritisation, resource allocation, and clinical decision support. Overall, findings suggest that AI systems can enhance triage accuracy, reduce waiting times, and improve operational efficiency in emergency care settings. However, the authors highlight persistent limitations, including algorithmic bias, limited interpretability, variability in data quality, ethical concerns, and a lack of prospective evaluation in real-world clinical environments.

3) (Araouchi & Adda, 2024)

(Araouchi & Adda, 2024, 2024) developed TriageIntelli, a multimodal AI-assisted triage system for healthcare settings, evaluated using multiple machine learning approaches, including Support Vector Machines, Random Forests, Neural Networks, Gradient Boosting Machines, XGBoost, and a stacking ensemble method. The models were trained and tested on a single structured dataset, with performance evaluated using classification accuracy, where the stacking ensemble achieved the highest score of 80.05%. Although the results indicate strong predictive performance under controlled conditions, the system was not validated in a live clinical environment or across multiple institutions, limiting its external validity and practical deployment potential.

4) (Almulihi et al., 2024)

(Almulihi et al., 2024) conducted a systematic review examining how artificial intelligence and machine learning can improve emergency medicine triage, particularly in patient prioritisation, admission prediction, and clinical outcome assessment. The authors reviewed studies that developed and evaluated predictive machine learning models for emergency department decision-support. Across the reviewed literature, AI systems consistently demonstrated higher predictive performance than traditional triage approaches and showed potential to improve resource allocation and patient management. However, most studies relied on retrospective datasets and lacked external validation. The review also identified concerns regarding algorithm transparency, bias, and integration into existing clinical workflows as key limitations to implementation.

5) (Tahernejad et al., 2024)

(Tahernejad et al., 2024) conducted a systematic review investigating the use of AI technologies in emergency and disaster triage, where rapid patient prioritisation and resource allocation are critical. The review examined studies applying AI methods to support clinical decision-making, patient classification, and management of limited healthcare resources during high-demand situations. The findings suggested that AI-assisted triage can improve efficiency, consistency, and allocation of resources compared with traditional approaches. However, the authors identified several limitations, including a lack of standardised datasets, interoperability challenges between healthcare systems, ethical concerns, and limited evidence from prospective real-world deployments. They concluded that further implementation studies are required before widespread adoption can be recommended.

6) (Feretzakis et al., 2024)

(Feretzakis et al., 2024) developed a Gradient Boosting Machine (GBM) model to predict emergency department (ED) patient disposition, specifically admission versus discharge. Using structured triage variables such as acuity, waiting time, age, and arrival mode, the model achieved an AUC of 0.8256 and an accuracy of 73.17%, demonstrating moderate predictive performance. However, the model showed relatively low sensitivity (53.08%), indicating that some patients requiring admission may be incorrectly classified. The authors highlighted limitations including the retrospective design, lack of external validation, and the potential for historical clinical biases to affect predictions.

7) (Freitas et al., 2020)

(Freitas et al., 2020) conducted a qualitative observational study investigating factors influencing patient flow within an emergency department in Trinidad and Tobago. Using process mapping and thematic analysis, the study found that staffing levels, physical layout, interdepartmental delays, and informal staff workarounds significantly affected patient movement and service efficiency. These organisational and structural factors contributed to bottlenecks and delays in care delivery. The authors noted that the single-site design limited the generalisability of the findings to other healthcare settings.

Literature and Identified Gaps

Gap 1: Lack of Prospective Evaluation of AI-Assisted Triage in Real-World Clinical Workflows

Despite consistent evidence that AI-assisted triage models achieve high predictive performance in retrospective and controlled studies (Tyler et al., 2024; Da'Costa et al., 2025), there is limited evaluation of how these systems perform when integrated into real emergency department workflows. Existing research rarely examines operational outcomes such as triage efficiency, triage time, or consistency under real-time clinical pressure. This limitation is particularly important because emergency departments are dynamic environments characterised by time constraints, fluctuating patient volumes, and incomplete clinical information-conditions not reflected in retrospective datasets. In addition, human factors such as clinician trust, cognitive load, and potential automation bias remain underexplored despite their influence on AI-assisted decision-making. This gap can be addressed through a 12-week prospective pilot study assessing triage accuracy, inter-rater agreement, workflow efficiency, and triage time while maintaining clinician oversight of final decisions.

Gap 2: Limited External Validation and Generalisability of AI Triage Models

A second key limitation is the restricted external validity of existing AI triage systems. Most models are developed and evaluated using single-centre or narrowly defined datasets, limiting confidence in their performance across different patient populations, clinical protocols, and healthcare settings (Araouchi & Adda, 2024; Tyler et al., 2024). Systematic reviews further highlight that variations in demographics, data quality, and triage practices can significantly affect model performance (Almulihi et al., 2024; Tahernejad et al., 2024). Although some multimodal and advanced machine learning systems have been proposed, few have undergone robust validation in new clinical environments, raising concerns about generalisability and real-world applicability. This reflects an implementation gap, where strong performance in controlled settings does not consistently translate into routine clinical practice. This gap can be partially addressed through a 12-week pilot evaluating the AI tool on a local patient population and comparing outputs with clinician triage decisions, providing preliminary evidence of external validity and contextual adaptability within a real emergency department setting.

Preliminary Proposal

Statement of the Problem

Emergency departments face rising patient demand, prolonged waiting times, and variability in triage decision-making. Although AI-assisted triage systems have demonstrated improved predictive performance compared with traditional methods, most evidence comes from retrospective studies with limited real-world evaluation. There remains insufficient understanding of how AI decision-support affects triage consistency and workflow efficiency in routine practice. This 12-week pilot will evaluate an AI-assisted triage tool to assess its impact on triage accuracy, consistency, and assessment time while maintaining clinician oversight of final decisions.

Previous Work

Research on AI-assisted emergency department triage consistently demonstrates that machine learning models can improve prediction of patient outcomes and support more efficient resource allocation. (Tyler et al., 2024) found that ensemble machine learning methods frequently outperform traditional triage systems when predicting admission, mortality, and resource utilisation outcomes. (Da'Costa et al., 2025) reported that AI-assisted triage can improve prioritisation, reduce waiting times, and enhance operational efficiency, while also identifying challenges related to bias and interpretability. (Araouchi & Adda, 2024) demonstrated the potential of multimodal AI systems through the development of TriageIntelli, achieving promising predictive performance in a controlled environment.

Similarly, (Almulihi et al., 2024) found that AI models consistently outperform conventional triage approaches in predictive tasks and may improve resource allocation within emergency departments. (Tahernejad et al., 2024) extended these findings to emergency and disaster settings, highlighting AI's ability to improve consistency and support resource management during periods of high demand. Despite these promising results, the literature consistently identifies limitations related to retrospective study designs, limited external validation, lack of prospective implementation studies, concerns regarding algorithmic bias, and challenges integrating AI systems into clinical workflows.

Proposed Solution

This project proposes a prospective observational pilot of an AI-assisted triage decision-support tool within an emergency department. The pilot is designed to address two recurring limitations identified in the literature: the lack of prospective evaluation in real-world clinical workflows and the limited external validation of AI triage systems. Rather than replacing clinician judgement, the system will provide decision-support recommendations while clinicians retain responsibility for all final triage decisions.

To address the first gap, the study will evaluate how AI recommendations influence triage consistency, workflow efficiency, and assessment time during routine emergency department operations. To address the second gap, the pilot will assess the performance of the AI tool on a local patient population and compare its recommendations with existing triage outcomes, providing preliminary evidence of its generalisability beyond the datasets on which similar systems were originally developed.

The study will evaluate:

- Agreement between AI recommendations and clinician-assigned triage levels.
- Triage consistency and inter-rater reliability.
- Time required to complete triage assessments.
- Practical challenges associated with implementation and adoption.

The findings will provide preliminary evidence regarding the feasibility, effectiveness, and operational impact of AI-assisted triage in routine clinical practice and help inform future larger-scale implementation studies.

Integration of AI Tools

AI tools (ChatGPT) were used to support the structuring, refinement, and academic editing of this literature review and proposal. Their use included improving clarity, ensuring a consistent academic tone, and assisting in the synthesis of recurring themes across studies, particularly limitations related to real-world validation and external generalisability. All analytical interpretations were independently reviewed, and no AI-generated content was used without manual verification and acceptance.

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